

Spring 2021

Problem 1 (Fa11, Sp21). Find the volume of the solid given by $x^2 + z^2 \leq 1$, $y^2 + z^2 \leq 1$.

Problem 2 (Sp06, Fa10, Sp13, Sp21). Prove that the intersection of a decreasing sequence of nonempty closed connected subsets of a compact metric space is nonempty and connected. Give an example to show that this is false if the assumption that the space is compact is dropped.

Problem 3 (Fa10, Sp21). Show that the series $\sum_{n=1}^{\infty} \sin \frac{x}{n^2}$ converges uniformly on any bounded interval in $\mathbb{R}$.

Problem 4 (Fa10, Sp21, Sp26). If f is an analytic function from the unit disc into itself with $f(0) = 0$, prove that $|f'(0)| \leq 1$ and find all functions f for which equality holds.

Problem 5 (Fa77, Su82, Fa97, Sp01, Fa03, Fa08, Sp11, Fa12, Sp18, Sp21). Evaluate $\int_{-\infty}^{\infty} \frac{dx}{1+x^{2n}}$, where n is a positive integer.

Problem 6 (Sp83, Fa11, Sp21). Let A be an $n \times n$ real matrix where all entries not on the diagonal are negative, and the sum of the entries in each column is positive. Show that the determinant of A is positive.

Problem 7 (Sp12, Sp21). Suppose L is a linear operator acting on a non trivial vector space V over a field $\mathbb{K}$. Suppose $p(x) \in \mathbb{K}[x]$ is not identically zero and $p(L) = 0$. Show every eigenvalue of L is a root of p . Show that if p factors completely over $\mathbb{K}$ then some roots of p are eigenvalues of L .

Problem 8 (Sp85, Sp11, Fa17, Sp21). Find an irreducible polynomial over the integers with $2 \cos(2\pi/7)$ as a zero, and use this to show that it is not contained in any extension of the rational numbers of degree a power of 2.

Problem 9 (Su80, Sp12, Sp21, Sp22, Fa22). Let G be a finite group and H be a subgroup.

1. Show that the number of subgroups of G conjugate to H divides the index of H .
2. Show that if $G = \bigcup_{g \in G} gHg^{-1}$ then $G = H$.

Problem 10 (Su80, Sp97, Fa11, Fa13, Fa17, Sp21, Fa25). For each $(a, b, c) \in \mathbb{R}^3$, consider the series

$$\sum_{n=3}^{\infty} \frac{a^n}{n^b \log^c n}$$

Determine the values of (a, b, c) for which the series converges absolutely; converges but not absolutely; diverges.

Problem 11 (Sp12, Sp21). Suppose that X is a compact metric space. If Y is another metric space (possibly noncompact), let $\pi : X \times Y \rightarrow Y$ be the map $\pi(x, y) = y$. Show that if Z is a closed subset of $X \times Y$ then $\pi(Z)$ is closed in Y .

Problem 12 (Fa12, Sp21). Prove the existence of the limit

$$\lim_{n \rightarrow \infty} \left(\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} - \log n \right)$$

Problem 13 (Sp11, Sp21). If $0 < r < 1$, find $\sum_{k=0}^{\infty} r^k \cos k\theta$.

Problem 14 (Fa11, Sp21). If a and b are points in the open unit disc of the complex plane, show that there is a holomorphic map from the open unit disc to itself with holomorphic inverse that takes a to b .

Problem 15 (Sp11, Sp21). For each of the following 4 statements, give either a counterexample or a reason why it is true.

1. For every real matrix A there is a real matrix B with $B^{-1}AB$ diagonal.
2. For every complex matrix A there is a complex matrix B with $B^{-1}AB$ diagonal.
3. For every symmetric complex matrix A there is a complex matrix B with $B^{-1}AB$ diagonal.
4. For every symmetric real matrix A there is a real matrix B with $B^{-1}AB$ diagonal.

Problem 16 (Sp12, Sp21). Find the eigenvalues of the $n \times n$ matrix A with entries a_{ij} , where

$$a_{ij} = \begin{cases} 1 & i = j + 1 \\ -1 & i = j - 1 \\ 0 & \text{otherwise} \end{cases}$$

Problem 17 (Fa12, Sp21). Does there exist a homomorphism of commutative rings with unit from $\mathbb{Z}[x]/\langle x^2 + 3 \rangle$ to $\mathbb{Z}[x]/\langle x^2 - x + 1 \rangle$? Either exhibit such a homomorphism, or prove that none exists.

Problem 18 (Sp82, Sp11, Fa17, Sp21). Prove that the polynomial $x^4 + x + 1$ is irreducible over $\mathbb{Q}$.